Chapter 15, Problem 32P

Problem

Determine the inverse Laplace transform of each of the following functions:

(a) $\frac{8(s+1)(s+3)}{s(s+2)(s+4)}$

(b) $\frac{s^2-2s+4}{(s+1)(s+2)^2}$

(c) $\frac{s^2+1}{(s+3)(s^2+4s+5)}$

Step-by-step solution

Step 1 of 9

(a) Given function

$$F(s)=\frac{8(s+1)(s+3)}{s(s+2)(s+4)}$$

Using partial fraction expansion

$$F(s)=\frac{A}{s}+\frac{B}{s+2}+\frac{C}{s+4}$$

Residue Method: $A=sF(s)|_{S=0}=\frac{8(0+1)(0+3)}{(0+2)(0+4)}$

$$A=sF(s)|_{S=0}=\frac{8(1)(3)}{(2)(4)}=3$$

$$B=(s+2)F(s)|_{S=-2}=\frac{8(-2+1)(-2+3)}{(-2)(-2+4)}$$

$$B=(s+2)F(s)|_{S=-2}=\frac{8(-1)(1)}{(-2)(2)}=2$$

$$C=(s+4)F(s)|_{s=-4}=\frac{8(-4+1)(-4+3)}{-4(-4+2)} \quad C=(s+4)F(s)|_{s=-4}=\frac{8(-3)(-1)}{-4(-2)}=3$$

Substituting A,B and C values in $F(s)$

$$F(s)=\frac{3}{s}+\frac{2}{s+2}+\frac{3}{s+4} \quad \text{................ (1)}$$

Applying inverse transform to equation (1)

$$L^{-1}[F(s)]=L^{-1}\left[\frac{3}{s}+\frac{2}{s+2}+\frac{3}{s+4}\right]$$

$$L^{-1}[F(s)]=L^{-1}\left[\frac{3}{s}\right]+L^{-1}\left[\frac{2}{s+2}\right]+L^{-1}\left[\frac{3}{s+2}\right]$$

$$\left(\because L^{-1}\left[\frac{1}{s+a}\right]=e^{-at} \text{ and } L^{-1}\left[\frac{n!}{s^{n+1}}\right]=t^n\right)$$

$$\boxed{f(t)=\boxed{\left[3+2e^{-2t}+3e^{-4t}\right]u(t)}}$$

Step 2 of 9

(b) Given function

$$F(s)=\frac{s^2-2s+4}{(s+1)(s+2)^2}$$

Using partial fraction expansion

$$F(s)=\frac{A}{s+1}+\frac{B}{(s+2)^2}+\frac{C}{(s+2)}$$

Residue Method:

$$A=(s+1)F(s)|_{s=-1}=\frac{(-1)^2-(2)(-1)+4}{(-1+2)^2}$$

$$A=(s+1)F(s)|_{s=-1}=\frac{1+2+4}{(1)^2}=7$$

$$B=(s+2)^2F(s)|_{s=-2}=\frac{(-2)^2-2(-2)+4}{(-2+1)}$$

$$B=(s+2)^2F(s)|_{s=-2}=\frac{4+4+4}{-1}=-12$$

$$C=\frac{d}{ds}\left[(s+2)^2F(s)\right]|_{s=-2}=\frac{d}{ds}\left(\frac{s^2-2s+4}{s+1}\right)|_{s=-2}$$

$$=\frac{(s+1)(2s-2)-1(s^2-2s+4)}{(s+1)^2}|_{s=-2}$$

$$C=\frac{(-1)(-4-2)-(4+4+4)}{(-2+1)^2}=\frac{6-12}{1}=-6$$

Step 3 of 9

Substitute values in $F(s)$

$$F(s)=\frac{7}{s+1}-\frac{12}{(s+2)^2}-\frac{6}{s+2}$$

Applying inverse transform for the above equation

$$L^{-1}[F(s)]=L^{-1}\left[\frac{7}{s+1}-\frac{12}{(s+2)^2}-\frac{6}{s+2}\right]$$

$$L^{-1}[F(s)]=L^{-1}\left[\frac{7}{s+1}\right]-L^{-1}\left[\frac{12}{(s+2)^2}\right]-L^{-1}\left[\frac{6}{s+2}\right]$$

$$f(t)=\boxed{\left(7e^{-t}-12te^{-2t}-6e^{-2t}\right)u(t)}\left(\because L^{-1}\left[\frac{1}{s+a}\right]=e^{-at}\text{and } L^{-1}\left[\frac{n!}{(s+a)^{n+1}}\right]=t^n e^{-at}\right)$$

(c) Given function

$$F(s)=\frac{s^2+1}{(s+3)(s^2+4s+5)}=\frac{A}{s+3}+\frac{Bs+C}{s^2+4s+5}$$

$$F(s)=\frac{A}{s+3}+\frac{Bs+C}{s^2+4s+5}$$

$$\frac{s^2+1}{(s+3)(s^2+4s+5)}=\frac{A(s^2+4s+5)+(Bs+C)(s+3)}{(s+3)(s^2+4s+5)}$$

$$s^2+1=A(s^2+4s+5)+(Bs+C)(s+3)$$

$$s^2+1=A(s^2+4s+5)+(Bs^2+3Bs+Cs+3C)$$

$$s^2+1=s^2(A+B)+s(4A+3B+C)+(5A+3C)$$ ……………..(A)

Comparing s^2 coefficient in equation (A)

$A+B=1$ …………….. (1)

Comparing s coefficient in equation (A)

$4A+3B+C=0$ ……………. (2)

Comparing constants in equation (A)

$5A+3C=1$ ……………. (3)

Step 4 of 9

From equation (1) we can write

$A=1-B$

Substituting $A=1-B$ in (2)

$$4(1-B)+3B+C=0$$

$$4-4B+3B+C=0$$

$4-B+C=0$ …………….. (4)

Step 5 of 9

Substituting $A=1-B$ in (3)

$$5(1-B)+3C=1$$

$$5-5B+3C=1$$

$-5B+3C+4=0$ ……………. (5)

Step 6 of 9

Multiplying equation (4) with 5 we get

$$[4-B+C]\times 5=0\times 5$$

$20-5B+5C=0$ ………….. (6)

Subtract equation (6) with (5) we get

$$20-5B+5C-(-5B+3C+4)=0$$

$$20-5B+5C+5B-3C-4=0$$

$$16+2C=0$$

$$2C=-16$$

$$C=\frac{-16}{2}$$

$C=-8$ ………………….. (7)

Step 7 of 9

Substitute $C=-8$ in (3)

$$5A+3(-8)=1$$

$$5A=1+24$$

$$5A=25$$

$$A=\frac{25}{5}$$

$$A=5$$

Step 8 of 9

Take $A=1-B$

$$5=1-B$$

$$B=1-5$$

$$B=-4$$

Step 9 of 9

$$F(s)=\frac{5}{s+3}+\frac{-4s-8}{s^2+4s+5}$$

$$F(s)=\frac{5}{s+3}+\frac{-4(s+2)}{(s+2)^2+1^2}$$

Taking inverse transform we can write

$$f(t)=\boxed{5e^{-3t}-4e^{-2t}(\cos t)u(t)}$$

$$\left(\because L^{-1}\left[\frac{1}{s+a}\right]=e^{-at}\text{ and }L^{-1}\left[\frac{s+a}{(s+a)^2+\omega^2}\right]=e^{-at}\cos\omega t\right)$$